

Design and Ethics Considerations

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Capstone Team #	12
Capstone Name	Viking Motorsports EV Battery Management System
Team members	Jose Ramirez, Hussein Khodor, David Fransis, Fangdong Yuan

Part 1: Design Considerations

You are asked to first consider these issues and then decide whether they are relevant to your capstone project. Specifically, you are asked to decide whether your design needs to consider:

- Public health, safety, and welfare
- Global factors
- Cultural factors
- Environmental factors
- Economic factors

It may be unlikely that your design should consider all of these, but it is very unlikely (or impossible) that it should not consider any of these. So, the first stage is for your team to consider each of these factors and decide if they are relevant to your design. Next, you should give a short explanation for each, especially for those that you think are relevant. Explain why they are relevant and how your design may address it. It is OK if your capstone doesn't address all of these issues. You might think about how the future of this project might affect these categories, if that helps.

Did you consider various factors in the design of your project/product? Choose Yes / No / Possibly, and discuss why you did / didn't / might have.

Consider:	Yes/No/Possibly	Why?
Public health, safety, and welfare	Yes	Our system's purpose is to ensure the safety of the vehicle and driver operating it during competition according to the FSAE.
Global factors	Yes	The design of our Cell Monitoring module works only for LFP battery cells which are mostly produced in foreign countries such as China.
Cultural factors	No	Cultural factors are not very relevant because this project is mainly a technical battery monitoring and protection system, not a consumer product based on cultural preferences.
Environmental factors	Yes	This is relevant because the project uses rechargeable battery cells, and a BMS can help extend battery life by preventing unsafe operating conditions like over or under voltage. Longer battery life reduces waste and supports safer use of electric power systems.
Economic factors	No	The LFP battery cells which our design is based around have been dropping in price which means that making battery modules and packs with them will not be as costly as before.

Part 2: Ethics Considerations

In industry, you will have to recognize your own and your company's ethical and professional responsibilities. Not only would you have to recognize them, but you would also have to make judgments regarding them. In this project, you are asked to only consider if there are any potential ethical problems that could come up in your device (product) but the quality of your explanations (i.e., your judgment) will not be evaluated. Please consider the impact of your capstone project in these contexts:

- Global
- Economic
- Environmental
- Societal

Did you consider various contexts in which ethical issues may arise in the design of your project/product? Choose if it's applicable / not applicable / might be applicable, and why / why not / why it might be.

Consider a context for ethical decision-making	Applicable: Yes/No/Possibly	Why?
Global	No	Our project will only be used in a competition among Universities and will not be mass produced for global distribution.
Economic	No	This is not a major concern because our project does not affect employment, markets or resources in any meaningful way. The cost of components matters for design viability, but it does not carry any major ethical issues.
Environmental	Yes	The mining for the Lithium used in the LFP cells used for our design causes CO ₂ emissions, heavy water consumption, and land disruption for the areas affected.
Societal	Yes	This is applicable because the system involves battery safety, and a failure could affect users or people nearby through overheating or shorts.